Differentiable Sensor Design for Wake-Based Flow Inference

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Scope of this technical note

The project README provides the main scientific motivation, Tesseract framework and "Why Tesseract?" discussion, headline results, reproducibility instructions, engineering notes, and repository organization. This document is intended as a technical companion, providing additional mathematical and methodological detail on the flow model, inverse problem, surrogate construction, sensor-design objective, optimization routes, limitations, and future research directions.

1 Physical problem and motivation

Many fluid-mechanical systems must be characterized from sparse measurements rather than from direct access to the full flow state. In wind-tunnel experiments, aerodynamic monitoring, and feedback-control applications, for example, only a small number of pressure or velocity measurements may be available. In such settings, inference quality depends not only on the number of sensors, but also on *where they are placed*. Poorly positioned sensors may provide little information or make physically distinct states appear nearly identical.

We study this question using a canonical separated-flow problem: a thin flat plate held at a fixed but unknown angle of attack α in a two-dimensional incompressible flow. We ask whether sparse downstream velocity measurements can recover the hidden AoA and whether the sensor locations can be optimized to make different candidate angles easier to distinguish.

The flow is governed by the nondimensional incompressible Navier–Stokes equations,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2\mathbf{u} + \int_r \mathbf{f}\big(\boldsymbol{\xi}(r,t)\big)\delta\big(\boldsymbol{\xi}-\mathbf{x}\big)\,dr, \tag{1}$$

$$\nabla\cdot\mathbf{u} = 0, \tag{2}$$

$$\mathbf{u}\big(\boldsymbol{\xi}(r,t)\big) = \int_{\mathbf{x}} \mathbf{u}(\mathbf{x})\delta\big(\mathbf{x}-\boldsymbol{\xi}\big)\,d\mathbf{x} = \mathbf{u}_B\big(\boldsymbol{\xi}(r,t)\big). \tag{3}$$

Here $\mathbf{u} = (u_x, u_y)$ is the velocity field, p is the pressure, and $\mathbf{f}$ is the immersed-boundary force. The third equation imposes no slip at the Lagrangian boundary points $\boldsymbol{\xi}(r,t)$. $\mathbf{f}$ acts as the associated Lagrange multiplier and is spread to the Eulerian momentum equation through the regularized delta function. The simulations are performed with `Immersa.jl` using an immersed-boundary projection formulation, in which the plate is represented independently of the fixed Cartesian fluid grid. This avoids remeshing as the angle of attack changes.

The calculations use $Re = 200$, $h = 0.05$, $\Delta t = 0.0025$, and $t_f = 20$. Virtual probes record (u_x, u_y) at $t = \{12.0, 13.3, 15.1, 17.4, 20.0\}$. For fixed sensor locations $\mathbf{s}$, these measurements define the inverse problem for α; in experimental design, the candidate AoAs are fixed and $\mathbf{s}$ is optimized to increase their distinguishability and reduce inverse ambiguity.

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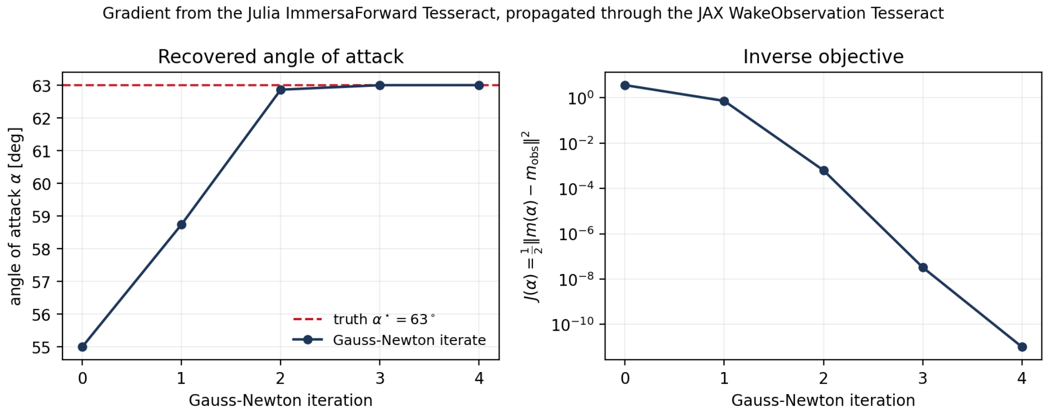

Figure 1: Hidden-AoA inference through the composed T1→T2 route. Left: Gauss–Newton iterates converge from $\alpha_0 = 55°$ to the hidden truth $\alpha^\star = 63°$. Right: the corresponding least-squares objective decreases by several orders of magnitude during the iteration.

2 Methodology and other results

The workflow is composed of four Tesseracts connected through three distinct computational routes: physical inverse inference, surrogate sensor design, and mechanistic sensor evaluation and refinement. This section describes each route mathematically, emphasizing the optimization variables, objectives, and derivative paths used in the study. For a higher-level overview of the workflow and its motivation, see the project `README`.

2.1 Physical inverse-inference route

We first consider the inverse problem with the sensor locations fixed. Let $\mathbf{s}$ denote a prescribed sensor layout and $\mathbf{m}(\alpha; \mathbf{s})$ the sparse wake measurements predicted for a candidate angle of attack α. These are obtained through the T1→T2 route: `ImmersaForward` computes the dense CFD fields, and `WakeObservation` samples them at the fixed probe locations and observation times.

Given observations $\mathbf{m}_{\text{obs}}$, define $\mathbf{r} = \mathbf{m}(\alpha; \mathbf{s}) - \mathbf{m}_{\text{obs}}$ and $\mathbf{J} = \partial \mathbf{m}/\partial \alpha$. The hidden AoA is recovered from the nonlinear least-squares problem

$$\min_{\alpha} \ \mathcal{L}_{\text{inv}} = \frac{1}{2}\mathbf{r}^T\mathbf{r}, \qquad \frac{d\mathcal{L}_{\text{inv}}}{d\alpha} = \mathbf{J}^T\mathbf{r}. \tag{4}$$

The measurement sensitivity $\mathbf{J}$ composes the T1 AoA sensitivity, computed by central finite differences inside the Julia component, with the downstream JAX derivative through T2. The inverse optimizer therefore sees the derivative of the complete measurement map without depending on how either component computes it.

Because α is scalar, we use a damped Gauss–Newton update,

$$\Delta\alpha_{\text{GN}} = -\frac{\mathbf{J}^T\mathbf{r}}{\mathbf{J}^T\mathbf{J}}, \tag{5}$$

with bounded steps and backtracking until the objective decreases. The search is restricted to $0° \leq \alpha \leq 90°$.

Figure 1 shows a representative recovery for a hidden truth $\alpha^\star = 63°$ starting from $\alpha_0 = 55°$. The composed T1→T2 workflow rapidly approaches the correct AoA while the inverse objective decreases by several orders of magnitude, converging to $\alpha = 63.000018°$.

2.2 Cheap sensor-design route

Repeated T1→T2 evaluations are too expensive for large-scale sensor optimization, so `WakeSurrogate` (T3) approximates the observable map

$$\text{T3}(\alpha, x, y) \approx (\text{T2} \circ \text{T1})(\alpha, x, y), \tag{6}$$

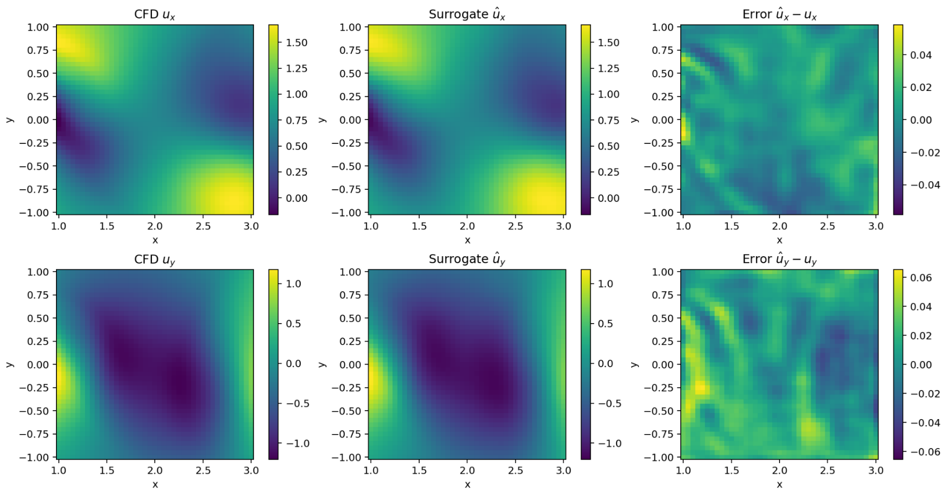

Figure 2: CFD observables (left), fields reconstructed from dense T3 coordinate queries (middle), and pointwise errors (right). During optimization, T3 is queried only at the probe locations.

using training data generated directly by the mechanistic pipeline. T3 models only the velocity observables required for sensor design, not the complete CFD state.

For each query (α, x, y), the coordinate-based model predicts

$$\widehat{\mathbf{u}}(\alpha, x, y) = [\hat{u}_x(t_1), \hat{u}_y(t_1), \ldots, \hat{u}_x(t_5), \hat{u}_y(t_5)]^T, \tag{7}$$

giving the two velocity components at that spatial location for all five observation times. During sensor optimization, T3 is evaluated only at the current probe coordinates; dense spatial queries are used only to reconstruct fields for validation.

Spatial coordinates are represented using Fourier features

$$\gamma(q) = [\sin(\pi f_k q), \cos(\pi f_k q)]_{f_k \in \{1,2,4,8\}}, \tag{8}$$

applied to x and y, while AoA remains a smooth scalar input. The resulting features are passed through a three-hidden-layer MLP of width 128 with SiLU activations. Training uses mean-squared error, Adam with learning rate 10^{-3}, batch size 512, and early stopping with patience 100 over at most 1000 epochs.

Figure 2 validates the learned coordinate map at an unseen α case. Dense queries of T3 closely reproduce the CFD velocity observables over the sensor-design region, with comparatively small pointwise errors.

The resulting measurements are passed to `SensorArrayDesign` (T4), which scores how informative the current layout is. For candidate AoAs α_i and α_j, with corresponding measurement vectors $\mathbf{m}_i(\mathbf{s})$ and $\mathbf{m}_j(\mathbf{s})$, T4 computes the normalized pairwise separation

$$d_{ij}(\mathbf{s}) = \frac{\|\mathbf{m}_i(\mathbf{s}) - \mathbf{m}_j(\mathbf{s})\|_2^2}{10 N_s}, \tag{9}$$

where $10N_s$ is the number of scalar measurements from N_s probes, two velocity components, and five observation times. Only pairs satisfying $|\alpha_i - \alpha_j| \geq 7.5^\circ$ are retained, so the objective focuses on physically distinct AoAs rather than being dominated by the naturally similar wakes of immediately neighboring angles.

T4 replaces the hard minimum over the retained pairs with a smooth soft-min objective $D_\tau(\mathbf{s})$, so larger values correspond to layouts whose most difficult-to-distinguish AoAs are better separated. Sensor design is posed as

$$\max_{\mathbf{s}} D_\tau(\mathbf{s}), \qquad (x_k, y_k) \in [1, 3] \times [-1, 1]. \tag{10}$$

The application-level bounded L-BFGS-B optimizer updates the sensor coordinates using reverse-mode derivatives through the composed T3→T4 route, while the candidate AoA grid remains fixed.

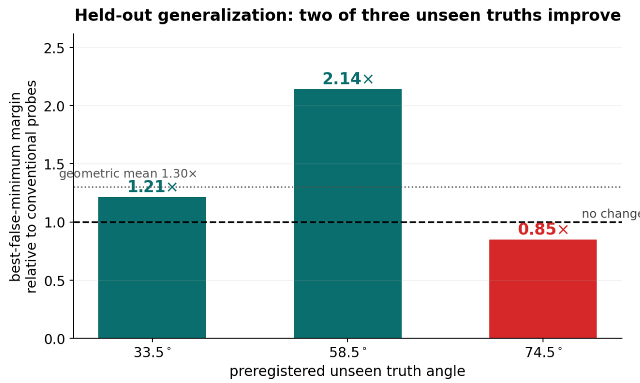

Figure 3: Held-out evaluation of the optimized sensor layout at $\alpha^\star = 33.5°$, $58.5°$, and $74.5°$, none of which belong to the design grid. The true-to-best-false margin changes by factors of approximately 1.213, 2.144, and 0.849, respectively, relative to the conventional layout. Thus, optimized sensing improves separation for two unseen truths but degrades it for one, indicating that generalization away from the design grid is not uniform.

Because the design landscape is nonconvex, each sensor budget is optimized from ten initial layouts: the conventional cross-stream rake and nine random starts.

2.3 High-fidelity sensor-design route

The same design objective can be evaluated directly on mechanistic CFD data through T1→T2→T4. Persisted T1 fields are sampled by T2 at the current probe coordinates and passed to T4. Because the probes are passive, sensor-coordinate derivatives act only through T2→T4.

This route both evaluates surrogate-designed layouts on the physical CFD fields and refines them further by continuing the optimization from the T3→T4 solution using the mechanistic objective and gradients. The quantitative comparison in the project `README` shows that this high-fidelity refinement moves the probes to a new layout and further improves the physical discrimination score relative to the surrogate-designed result.

3 Limitations

The discrimination objective is evaluated on a finite AoA design grid and is therefore a proxy, rather than a guarantee, for performance at unseen flow states. Figure 3 illustrates this limitation using three held-out truths. The true-to-best-false margin changes by factors of approximately 1.213, 2.144, and 0.849 at $\alpha^\star = 33.5°$, $58.5°$, and $74.5°$, respectively. Thus, optimized sensing improves separation for two unseen cases but degrades it for one, showing that generalization away from the design grid is not uniform.

The present study also uses deterministic measurements at five fixed observation times, while T1 currently obtains AoA sensitivities through finite differences.

4 Future work

A natural next step is to develop sensor-design objectives that account more directly for global inverse recoverability. One direction is alias- or basin-aware design, in which sensor locations are optimized to suppress false minima in the inverse landscape rather than only maximize finite-grid pairwise discrimination. A second direction is to differentiate through the inverse solver itself, so that layouts can be optimized for successful recovery over distributions of truths and initial guesses. Adaptive or sequential sensing could further use measurements already acquired to determine where or when subsequent observations should be taken. Finally, replacing the finite-difference AoA sensitivity inside T1 with a tangent-linear, JVP, or adjoint implementation would reduce the cost of mechanistic derivatives while preserving the same downstream Tesseract interface.